Manisudhan

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Professional Summary

Results-driven Computer Science engineering student specializing in Backend Development, DevOps, and Cloud Architecture. Proven ability to architect scalable systems, deploy cloud-native applications on AWS, and automate CI/CD pipelines. Proficient in Java, Python, and Node.js, with hands-on experience building RESTful APIs, integrating Computer Vision models, and managing relational databases. Strong focus on writing clean, maintainable code, leveraging containerization with Docker, and optimizing cloud infrastructure.

Technical Skills

Programming Languages: Java, Python, Node.js (JavaScript), SQL, C++, HTML/CSS

Backend & Cloud: AWS (EC2, S3, Application Load Balancer, VPC), Express.js, RESTful APIs

DevOps & Tools: Linux, Docker, Git, GitHub Actions, CI/CD Automation

Databases: PostgreSQL, Supabase, IndexedDB, MySQL

Core Competencies: Data Structures & Algorithms (DSA), System Design, Computer Vision (YOLOv8), JWT Authentication

Education

SSN School of Engineering **2023–2027**
B.Tech in Computer Science and Engineering
Currently in 4th Year. Cumulative Grade Point Average (CGPA): 7.6/10.0

Projects

AI-Powered Automated Traffic Violation System **2026**

- Engineered an intelligent computer vision pipeline utilizing YOLOv8 and OpenCV to automate traffic monitoring and violation detection in real time.
- Developed and trained a custom deep learning model to accurately detect cabin-level details, specifically identifying whether a driver is wearing a seatbelt.
- Processed video streams to optimize frame-rate inference, structuring object detection outputs for seamless backend logging and analytics.
- **Tech Stack:** Python, YOLOv8, OpenCV, Deep Learning, Computer Vision.

Automated CI/CD Deployment Pipeline **2026**

- Developed an automated deployment pipeline linking GitHub repositories directly to AWS EC2 instances to streamline development workflows.
- Configured webhooks and custom Linux bash scripts to listen for codebase pushes, automatically pulling the latest code and restarting the server, ensuring zero-downtime updates.
- **Tech Stack:** AWS EC2, Linux/Bash, Git, Webhooks, Node.js.

NoteNest - Productivity Chrome Extension (Chrome Web Store) **2026**

- Engineered a side-panel browser extension for organized note-taking with nested folder structures.
- Implemented IndexedDB for optimized, persistent local storage, ensuring fast data retrieval without server-side latency.
- Successfully published to the Chrome Web Store, organically acquiring over 200+ active downloads.
- **Tech Stack:** Node.js, JavaScript, IndexedDB, HTML/CSS.

Cloud-Hosted Fitness Tracker Platform (workk.digital) **2025**

- Developed a full-stack fitness tracking web application with robust user authentication utilizing Google OAuth and JWT tokens.
- Designed and implemented RESTful APIs for CRUD operations, allowing users to seamlessly log, track, and manage daily workout routines.
- Deployed the scalable backend infrastructure on AWS EC2, ensuring high availability and secure data transmission.
- **Tech Stack:** Node.js, Express, AWS (EC2), JWT, OAuth, PostgreSQL.

Certifications & Experience

AWS Certified Solutions Architect – Associate – Amazon Web Services (AWS) **2026** hyperref
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- Earned formal validation for designing distributed, fault-tolerant, and scalable systems on AWS.
- Achieved a credential score of **820/1000**, demonstrating proficiency in cloud security, VPC architecture, compute services, and resilient storage strategies.

Virtual Internship Completion Certificate – Infosys Springboard **2025**

- Gained hands-on experience with industry-standard software development lifecycles and enterprise backend implementations.